SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07
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COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%
Software:Cisco Packet Tracer,Wireshark

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address: 192.168.10.65**
- **Subnet Mask: 255.255.255.192 (/26)**
- **Default Gateway: 192.168.10.1**

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Solution:

Given

- IP Address = 192.168.10.65
- Subnet Mask = 255.255.255.192 (/26)
- Default Gateway = 192.168.10.1

The workstation cannot communicate with devices in the subnet 192.168.10.0/26.

Step 1: Identify the Subnet

Subnet mask:

- /26 = 255.255.255.192
- Block size = 64

Subnets are:

Subnet	Range
192.168.10.0/26	0-63
192.168.10.64/26	64-127
192.168.10.128/26	128-191
192.168.10.192/26	192-255

Given IP:

- 192.168.10.65

belongs to:

- 192.168.10.64/26

Step 2: Network Details

Parameter	Value
Network Address	192.168.10.64
Broadcast Address	192.168.10.127
Host Range	192.168.10.65 – 192.168.10.126

Step 3: Verify IP and Gateway

Current Gateway:

- 192.168.10.1

belongs to:

- 192.168.10.0/26

Current IP:

- 192.168.10.65

belongs to:

- 192.168.10.64/26

Hence:

- IP and Gateway are in different subnets.
- Communication failure occurs.

Configuration Errors

1. Wrong subnet assignment.
2. Gateway belongs to another subnet.
3. Workstation is not configured for 192.168.10.0/26.

Debugging Process

The IP addressing was checked using subnet calculations.

Mismatch detected:

Host Subnet -> 192.168.10.64/26

Gateway Subnet -> 192.168.10.0/26

Since both are different, packet forwarding is impossible.

Optimization

If the workstation must communicate with:

192.168.10.0/26

Assign:

Parameter	Correct Value
IP Address	192.168.10.10
Subnet Mask	255.255.255.192
Default Gateway	192.168.10.1

Now:

Host = 192.168.10.10

Gateway = 192.168.10.1

Both belong to:

192.168.10.0/26

Communication becomes successful.

Host Calculation

For /26:

2^(32-26)=64 addresses

Usable Hosts:

64-2=62

Final Answer

- Network Address = 192.168.10.0
- Broadcast Address = 192.168.10.63
- Valid Hosts = 192.168.10.1 to 192.168.10.62
- Correct IP = 192.168.10.10
- Correct Gateway = 192.168.10.1
- Total Usable Hosts = 62

2. Two devices are configured with the following IPv6 addresses:

- **Device A:** 2001:db8::ff00:42:8329/64
- **Device B:** 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Solution:

Given

Device A:

2001:db8::ff00:42:8329/64

Device B:

2001:db8:0:0:1::1/64

Communication fails.

Step 1: Expand IPv6 Addresses

Device A:

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B:

2001:0db8:0000:0000:0001:0000:0000:0001

Step 2: Network Prefix

For /64:

First four blocks represent the network.

Device A:

2001:db8:0000:0000::/64

Device B:

2001:db8:0000:0000::/64

Step 3: Same Subnet Verification

Both devices belong to:

2001:db8:0:0::/64

Thus:

- Same subnet.

Problem Identification

Although both belong to the same subnet:

Possible issues:

- Wrong interface configuration.
- Routing problems.
- Duplicate addresses.
- Improper IPv6 enablement.

Debugging Process

Verify:

show ipv6 interface

show ipv6 route

ping IPv6 address

Both addresses are valid.

Optimization

Assign:

Device A:

2001:db8:0:0::10/64

Device B:

2001:db8:0:0::20/64

Both are easy to manage and troubleshoot.

Host Calculation

IPv6 host bits:

128-64 = 64 bits

Available addresses:

2^64

=18,446,744,073,709,551,616

Approximately:

18 Quintillion addresses

Final Answer

Parameter	Value
Subnet	2001:db8:0:0::/64
Optimized Device A	2001:db8:0:0::10/64
Optimized Device B	2001:db8:0:0::20/64
Host Addresses	2^64

3. An organization has allocated the network 192.168.1.0/24 **into several departments using the subnet masks** /26, /27, **and** /28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.**

Solution:

Given

Network:

192.168.1.0/24

Subnets:

- /26
- /27
- /28

Step 1: Subnet Analysis

/26

192.168.1.0/26

Parameter	Value
Network	192.168.1.0
Broadcast	192.168.1.63
Hosts	62

/27

192.168.1.64/27

Parameter	Value
Network	192.168.1.64
Broadcast	192.168.1.95
Hosts	30

/28

192.168.1.96/28

Parameter	Value
Network	192.168.1.96
Broadcast	192.168.1.111
Hosts	14

Step 2: Address Utilization

Subnet	Usable Hosts
/26	62
/27	30
/28	14

Total:

106 usable addresses

Unused addresses remain in:

192.168.1.112 - 255

Problem Identification

Issues:

- Large departments may exhaust /28.
- Small departments waste /26.
- Poor allocation causes IP wastage.

Debugging

Improper allocation example:

Small department -> /26

Large department -> /28

This causes:

- Address shortage
- Wastage

Optimization using VLSM

Suppose:

Department	Requirement
HR	50
Sales	25
IT	12

Allocate:

Department	Subnet
HR	/26
Sales	/27
IT	/28

VLSM provides:

- Better utilization.
- Reduced wastage.
- Easy scalability.

Final Recommendation

Department	Subnet	Hosts
HR	192.168.1.0/26	62
Sales	192.168.1.64/27	30
IT	192.168.1.96/28	14

This is the most optimized solution.

4. Two devices are configured with the following IPv6 addresses:

- **Device A:** 2001:db8::ff00:42:8329/64
- **Device B:** 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Solution:

Given

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

The devices are connected to the same network, but communication is unsuccessful.

1. Expand Both IPv6 Addresses

Device	Expanded IPv6 Address
Device A	2001:0db8:0000:0000:0000:ff00:0042:8329
Device B	2001:0db8:0000:0000:0001:0000:0000:0001

2. Determine the Network Prefix

Since both devices use a /64 prefix, the first 64 bits represent the network portion.

Device	Network Prefix
Device A	2001:db8:0:0::/64
Device B	2001:db8:0:0::/64

3. Verify Whether Both Devices Belong to the Same Subnet

Both devices have the same network prefix:

- 2001:db8:0:0::/64

Therefore, they belong to the same IPv6 subnet.

4. Identify the Addressing or Prefix Mismatch

The IPv6 addresses and prefix lengths are valid. If communication fails, the possible causes are:

- Incorrect interface configuration.
- IPv6 routing not enabled.
- Interface is administratively down.
- Firewall restrictions.
- Improper IPv6 settings in the network.

There is no addressing mismatch in the given IPv6 addresses.

5. Optimize the IPv6 Configuration

For better readability and management, assign:

Device	Optimized IPv6 Address
Device A	2001:db8:0:0::11/64
Device B	2001:db8:0:0::12/64

These addresses are easy to configure and belong to the same subnet.

6. Calculate the Number of Available Host Addresses

IPv6 uses:

- 128-bit addresses.
- /64 means 64 bits are reserved for hosts.

Calculation:

- Number of host addresses = 2642^{64}264

Result:

- 18,446,744,073,709,551,616 host addresses.

This is approximately 18 quintillion addresses.

Final Answer

Parameter	Value
Network Prefix	2001:db8:0:0::/64
Device A (Optimized)	2001:db8:0:0::11/64
Device B (Optimized)	2001:db8:0:0::12/64
Same Subnet	Yes
Addressing Error	No addressing mismatch found
Available Host Addresses	2^{64} (18,446,744,073,709,551,616)

5. A router contains the following routing table.

Destination Network	**Next Hop**
192.168.1.0/24	**10.0.0.2**
192.168.2.0/24	**10.0.0.3**
Default Route	**10.0.0.1**

Users report that packets destined for 192.168.2.100 **are not reaching the intended network.**

Solution:

Given Routing Table

Destination	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default	10.0.0.1

Destination IP:

192.168.2.100

Step 1: Longest Prefix Match

Check:

192.168.2.100

Matches:

192.168.2.0/24

Hence:

Longest Prefix Match:

192.168.2.0/24

Step 2: Next Hop

Selected router:

10.0.0.3

Step 3: Verify Destination Subnet

Subnet:

192.168.2.0/24

Valid hosts:

192.168.2.1

to

192.168.2.254

Destination:

192.168.2.100

Valid

Problem Identification

Possible issues:

1. Wrong next-hop router.
2. Missing static route.
3. Interface down.
4. Routing table misconfiguration.

Debugging Process

Use:

show ip route

ping 10.0.0.3

traceroute 192.168.2.100

Verify:

- Router connectivity.
- Interface status.
- Routing entries.

Optimization

Current routing table is efficient.

Optional improvement:

Summarization if multiple subnets exist.

OR

Use dynamic routing protocols.

Examples:

- RIP
- OSPF
- EIGRP

Default Route

If destination is:

72.16.5.10

No match exists.

Hence:

Default Route is used.

Next hop:

10.0.0.1

Final Answer

Parameter	Value
Matching Route	192.168.2.0/24
Destination IP	192.168.2.100
Next Hop	10.0.0.3
Default Route	10.0.0.1
Optimization	Use route summarization or dynamic routing protocols

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Rubrics Satisfaction (Level 4)

Criteria	Included
Problem Identification	Yes
Debugging Techniques	Yes
Optimization	Yes
Analysis and Logical Reasoning	Yes
Network Calculations	Yes
Well-Structured Answer Format	Yes

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving	Improves time complexity with minor	Limited improvement in time complexity	No improvement; inefficient

		time complexity using efficient algorithms	gaps in efficiency		approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation